

Thenmozhi Boopathy

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EDUCATION

Miami University Master of Science in Business Analytics	Oxford, OH Expected Graduation 2027
Mother Teresa Women's University SLET (State Level Eligibility Test), Mathematics	Kodaikanal, India 2017
Bharathidasan University - Master of Philosophy in Mathematics - Master of Science in Mathematics - Bachelor of Science in Mathematics	TamilNadu, India 2008 2007 2005

RELEVANT COURSEWORK

Data Mining: Exploratory Data Analysis, Dimension Reduction, Tree-Based Models, Logistic Regression, Neural Networks, Model Evaluation, **Big Data:** Core Python, Pandas, NoSQL (MongoDB), Web APIs, Scikit-learn.

PROJECT EXPERIENCE

Enterprise RAG Chatbot — Job Finder Assistant (ISA-632)	2026
Built and deployed an enterprise-grade Retrieval-Augmented Generation (RAG) chatbot on Databricks with a 3-person team: used Apache Spark to ingest 2,500+ job records into Delta Tables, integrated Databricks Vector Search with a LangChain agent on Llama 4 Maverick, and built a Streamlit UI with MLflow-tracked evaluation, achieving 89% positive user feedback. Live: analystjobfinder.streamlit.app	
End-to-End R Machine Learning Workflow for Loan Default Prediction (ISA-591)	2025
Designed and implemented an end-to-end R machine learning pipeline for loan-default prediction, engineering features, training multiple classification models, and evaluating performance with a focus on minimizing high-risk False Negatives.	
Theme Park Attendance Forecasting Pipeline with Python (ISA-514)	2025
Built a Python-based machine learning pipeline using MongoDB for data storage and integration of structured and unstructured sources (weather, wait times, schedules, Reddit sentiment) to forecast Six Flags attendance for operational decision-making.	
Inventory Models Dissertation - Graduate Project	2008
Built and analyzed deterministic inventory models to reduce costs, evaluate price-break impacts, and assess system stability under shortages.	

PROFESSIONAL EXPERIENCE

S.A College of Engineering <i>Assistant Professor</i>	Chennai, India 2015-2015
Delivered instruction in quantitative and statistical methods (numerical analysis, probability, matrices, and applied functions), emphasizing analytical problem-solving skills relevant to data-driven business decision-making.	
Sri Venkateswara College of Engineering <i>Assistant Professor</i>	Chennai, India 2012-2013
Performed lecturing in advanced quantitative methods (calculus, sequences, integrals), building strong analytical and problem-solving foundations relevant to data modeling and forecasting.	
Bon Secours College for Women <i>Lecturer (PG and UG)</i>	Tanjore, India 2009-2012
Conducted training in mathematics and analytics concepts for undergraduate and postgraduate students, strengthening quantitative reasoning, data interpretation, and problem-solving skills.	

VOLUNTEER ACTIVITIES

Language Teacher <i>Greater Cincinnati Tamil School, Cincinnati</i>	Cincinnati, Ohio 2021-2025
Designed and provided customized language programs for students of varying proficiency levels, using assessments to track progress and enhance outcomes. Coordinated program logistics and engaged stakeholders to promote the Seal of Bilingualism.	
Gifted Education Co-Representative <i>Sycamore Gear</i>	Cincinnati, Ohio 2023-2024
Designed and executed student engagement initiatives, enhancing community involvement and program effectiveness.	
Exam Proctor <i>Math Kangaroo USA, NFP</i>	Cincinnati, Ohio 2024-2025
Ensured exam integrity and provided guidance to students during online assessments.	

SKILLS

Certification: Databricks University Alliance Student — Databricks (issued 2026).

- Python - Jupyter Notebook - NoSQL - Visual Studio Code - Databricks - Apache Spark - GenAI / LLM	- Pandas - R - Power Bi - ML pipeline - LangChain - MLflow	- SQL - RStudio - GIT - Delta Lake - Vector Search
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